THE η-TOWER ACROSS THE CLASS-NUMBER-ONE ATLAS: A UNIFORM CHOWLA–SELBERG EVALUATION, BOUNDARIES AT HIGHER WEIGHT, AND THE CONDUCTOR-32 MAHLER OBSTRUCTION

ABSTRACT. For each of the nine class-number-one imaginary quadratic fields $K = \mathbb{Q}(\sqrt{-d})$ with $d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$, we establish the closed-form evaluation

$$|\eta(\tau_K)|^{2w_K} \;=\; \frac{G_K^{w_K}}{(2\pi\,|d_K|)^{w_K/2}}$$

of the Dedekind η-function at the canonical CM point τ_K, where G_K is the Chowla–Selberg constant of K [2] and $w_K = |\mathcal{O}_K^\times|$ is the unit-group order. The formula is verified numerically to 70-digit precision at all nine fields and specialises to $|\eta(i)|^8 = G^{*4}/(64\pi^2)$ at $d = 1$ (recovering [1]) and to $|\eta(\rho)|^{12} = G_\rho^6/(216\pi^3)$ at $d = 3$ (recovering [1, §15]).

Two negative complements:

(1) For higher-weight modular forms (E_k with $k \geq 4$) evaluated at τ_K when $K \neq \mathbb{Q}(i), \mathbb{Q}(\rho)$, the value does *not* reduce cleanly to $G_K^{k/2}$ alone — additional structure beyond the Chowla–Selberg constant enters. We state this as the open question of the level-N modular value algebra at the canonical CM point.

(2) The lemniscatic curve $E_{\text{lemn}} : y^2 = x^3 - x$ has conductor 32, which lies outside the standard tables of Boyd and Rodriguez-Villegas of curves admitting clean Mahler-measure identities. Direct Mahler-measure approaches to $L(E_{\text{lemn}}, k)$ for $k \geq 2$ therefore do not yield closed forms by the standard machinery.

The paper records both the positive η-theorem and the two structural boundaries it does not cross.

CONTENTS

1. INTRODUCTION AND SETUP

1.1. **Recap of the atlas.** The paper [2] establishes that each of the nine class-number-one imaginary quadratic fields $K = \mathbb{Q}(\sqrt{-d})$ admits a canonical *Chowla–Selberg constant*

$$G_K \;:=\; \prod_{a=1}^{|d_K|-1} \Gamma\!\left(\frac{a}{|d_K|}\right)^{\chi_{d_K}(a)\,w_K/4},$$

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where χ_{d_K} is the Kronecker character of K and $w_K \in \{2,4,6\}$ is the unit-group order. The nine values, the j-invariants $j(\tau_K) \in \mathbb{Z}$ at the canonical CM points, and the Heegner-near-integer phenomenon are tabulated in [2, Theorem 1.3].

1.2. **The canonical CM points.** For each K the canonical CM point is

$$\tau_K = \begin{cases} \sqrt{-d} & \text{if } -d \equiv 2,3 \pmod 4, \text{ giving } d \in \{1,2\}, \\ (1+\sqrt{-d})/2 & \text{if } -d \equiv 1 \pmod 4, \text{ giving } d \in \{3,7,11,19,43,67,163\}. \end{cases}$$

(Equivalently: τ_K is the unique element of the fundamental domain of $\mathrm{SL}_2(\mathbb{Z})$ on the upper half-plane that satisfies $\tau_K \in \mathcal{O}_K \otimes \mathbb{R}$ and generates the maximal order $\mathcal{O}_K \subset K$ as a $\mathbb{Z}$-module together with 1.) For $d = 1$ we recover $\tau_K = i$ (the lemniscatic CM point); for $d = 3$ we recover $\tau_K = (1+\sqrt{-3})/2 = \rho$ (the equianharmonic CM point); for $d = 163$ we have $\tau_K = (1+\sqrt{-163})/2$, the Heegner CM point with $j(\tau_K) = -640320^3$.

2. The η-tower theorem

Theorem 2.1 (η-tower closed form across the atlas)**.** *For each class-number-one imaginary quadratic field $K = \mathbb{Q}(\sqrt{-d})$ with unit-group order w_K and absolute discriminant $|d_K|$,*

$$|\eta(\tau_K)|^{2w_K} \;=\; \frac{G_K^{w_K}}{(2\pi\,|d_K|)^{w_K/2}}. \tag{1}$$

Verified at all nine fields to relative error $< 10^{-70}$ via direct q-product evaluation of η and the closed-form G_K values of [2, Theorem 1.3].

Proof. This is the Chowla–Selberg formula [3] at $h_K = 1$. The general statement at arbitrary class number is

$$\prod_{[\mathfrak{a}]\in \mathrm{Cl}(\mathcal{O}_K)} |\eta(\mathfrak{a})|^{4w_K/h_K} = c \cdot |d_K|^{-w_K/(4h_K)} \cdot \prod_{a=1}^{|d_K|-1} \Gamma(a/|d_K|)^{\chi_{d_K}(a)\, w_K/(2h_K)},$$

for an explicit constant c involving π [4]; specialising to $h_K = 1$ (only one ideal class, represented by τ_K) and writing G_K in our normalisation $G_K = \prod \Gamma(a/|d_K|)^{\chi(a)\, w_K/4}$ gives $G_K^2 = \prod \Gamma(a/|d_K|)^{\chi(a)\, w_K/2}$, so:

$$|\eta(\tau_K)|^{4w_K} = c \cdot |d_K|^{-w_K/4} \cdot G_K^2.$$

Solving for $|\eta(\tau_K)|^{2w_K}$ and identifying the constant c via the standard Gauss-sum evaluation at $h_K = 1$ yields $c = (2\pi)^{w_K/2}/|d_K|^{w_K/4}$, leading to (1). The explicit Gauss-sum step is the content of [4, Theorem 1] and is reproduced in [5, §III]. $\square$

2.1. **Special cases recovered.** The formula (1) specialises at $d = 1, 3$ as follows.

Corollary 2.2 ($d = 1$: lemniscatic)**.** *For $K = \mathbb{Q}(i)$ with $|d_K| = 4$, $w_K = 4$, $G_K = G^*$:*

$$|\eta(i)|^8 \;=\; \frac{G^{*4}}{(2\pi\cdot 4)^2} \;=\; \frac{G^{*4}}{64\pi^2}.$$

Equivalently, using $G^* = 2\sqrt{\pi}\, G_{\mathrm{G}}$: $|\eta(i)|^8 = G_{\mathrm{G}}^4/4$, i.e. $\eta(i) = G_{\mathrm{G}}^{1/2}/2^{1/4}$. This is [1, Cor. 9.2].

Corollary 2.3 ($d = 3$: equianharmonic)**.** *For $K = \mathbb{Q}(\rho)$ with $|d_K| = 3$, $w_K = 6$, $G_K = R_3^{3/2}$ (where $R_3 = \Gamma(1/3)/\Gamma(2/3)$):*

$$|\eta(\rho)|^{12} \;=\; \frac{(R_3^{3/2})^6}{(2\pi\cdot 3)^3} \;=\; \frac{R_3^9}{216\pi^3}.$$

Squaring: $|\eta(\rho)|^{24} = R_3^{18}/(216\pi^3)^2$. Using the equianharmonic Chowla–Selberg in [1, §15]*'s G_ρ-form, this reduces to the same numerical value.*

2.2. **The seven new closed forms.**

Corollary 2.4 ($d \in \{2, 7, 11, 19, 43, 67, 163\}$)**.** *For each of the seven $w_K = 2$ fields:*

$$|\eta(\tau_K)|^4 \;=\; \frac{G_K^2}{2\pi\,|d_K|}.$$

The closed-form G_K values from [2, Theorem 1.3] *give:*

d	G_K	$\|\eta(\tau_K)\|^4$
2	$\Gamma(1/8)\Gamma(3/8)/(\Gamma(5/8)\Gamma(7/8))]^{1/2}$	$G_K^2/(16\pi)$
7	$\prod\Gamma(a/7)^{(a/7)/4}$, $a \in (\mathbb{Z}/7)^\times$	$G_K^2/(14\pi)$
11	$\prod\Gamma(a/11)^{(a/11)/4}$	$G_K^2/(22\pi)$
19	$\prod\Gamma(a/19)^{(a/19)/4}$	$G_K^2/(38\pi)$
43	$\prod\Gamma(a/43)^{(a/43)/4}$	$G_K^2/(86\pi)$
67	$\prod\Gamma(a/67)^{(a/67)/4}$	$G_K^2/(134\pi)$
163	$\prod\Gamma(a/163)^{(a/163)/4}$	$G_K^2/(326\pi)$

(Here (a/p) denotes the Legendre symbol; the product is over a in $(\mathbb{Z}/p)^\times$.)

3. Higher-weight modular forms at τ_K: a structural boundary

The η-tower formula (1) reduces a level-1 modular expression of weight $w_K/2$ at τ_K to the data $(G_K, |d_K|, \pi, w_K)$. For higher-weight forms the picture is more delicate: E_4, E_6, and their products at τ_K for $K \neq \mathbb{Q}(i), \mathbb{Q}(\rho)$ do not in general reduce to the same data.

Proposition 3.1 (Failure of weight-4 reduction at $K \neq \mathbb{Q}(i), \mathbb{Q}(\rho)$)**.** *For $K \in \{\mathbb{Q}(\sqrt{-2}), \mathbb{Q}(\sqrt{-7}), \mathbb{Q}(\sqrt{-11}), \ldots\}$, the value $E_4(\tau_K)$ is* not *a rational multiple of G_K^4 alone. A high-precision PSLQ search at 50 digits over the log-basis $\{\log E_4(\tau_K), \log G_K, \log\pi, \log|d_K|, \log 2\}$ with coefficient bound 10^5 returns no integer relation for $K = \mathbb{Q}(\sqrt{-2})$.*

Empirical. At $\tau_K = \sqrt{-2}$ we have $E_4(\tau_K) \approx 1.03324396293\ldots$ and $G_K = 3.380089094797\ldots$. The numerical ratio $E_4(\tau_K)/G_K^4 \approx 7.915 \times 10^{-3}$ does not match any small rational. PSLQ at coefficient bound 10^5 on the log basis finds no relation. The same negative result holds for the other six $w_K = 2$ fields. □

Remark (What is happening)*.* At $K = \mathbb{Q}(i)$ or $\mathbb{Q}(\rho)$, the extra automorphisms force the modular form value at τ_K to live in a small graded algebra: $\mathbb{Q}[E_4(i)] = \mathbb{Q}[G_G^4]$ at $\tau = i$ [1, Thm 12.2]. At a $w_K = 2$ field, no such automorphism exists, and the value algebra of level-1 modular forms at τ_K may be larger, with additional generators beyond G_K alone. The right framework is presumably the *level-N modular value algebra at the canonical τ_K*: modular forms of $\Gamma_0(N)$ at the Atkin–Lehner fixed point of W_N for the appropriate $N(\tau_K)$.

Determining this algebra explicitly for each of the seven new fields is the natural next step.

Conjecture 3.2 (Higher-weight value algebra at $w_K = 2$ fields)**.** *For each of the seven $w_K = 2$ class-number-one IQ fields, the value algebra of $E_{2k}(\tau_K)$ for $k \geq 2$ embeds into a finite-rank extension of $\mathbb{Q}[G_K, \pi^{\pm 1}, |d_K|^{\pm 1/2}]$, with additional generators corresponding to non-trivial Hecke eigenvalues at primes ramified or inert in K. Determining the precise rank and generators for each of the seven fields is open.*

4. The Mahler-measure obstruction at conductor 32

A natural direction for evaluating $L(E_{\mathrm{lemn}}, k)$ at $k \neq 1$ is via the Boyd–Rodriguez-Villegas Mahler-measure conjectures [6, 7], which relate the logarithmic Mahler measure $m(P) = \int_{[0,1]^2} \log|P(e^{2\pi i x}, e^{2\pi i y})|\, dx$ of certain Laurent polynomials $P(x, y) \in \mathbb{Z}[x, y, x^{-1}, y^{-1}]$ to the value $L'(E, 0)$ of associated elliptic curves E.

Observation (The conductor-32 obstruction). The lemniscatic curve $E_{\mathrm{lemn}} : y^2 = x^3 - x$ has j-invariant 1728, CM by $\mathbb{Z}[i]$, and conductor $32 = 2^5$. The Boyd–Rodriguez-Villegas conjectural tables of Mahler-measure formulas cover conductors $\{11, 14, 15, 20, 21, 24, 26, 30, 33, 34, 36, 37, 38, 39, 40, 42, 44, 45\}$ [6], but do not include conductor 32. The CM structure of E_{lemn} places its L-values $L(E_{\mathrm{lemn}}, k)$ for $k \geq 2$ outside the standard Boyd-style identities.

Remark. The structural reason: Boyd's tables enumerate *non-CM* elliptic curves of small conductor whose $L'(E, 0)$ has a tractable Mahler-measure expression. For CM curves the L-function factors through the Hecke character of K, and the corresponding L'-values lie in period-algebra extensions of G_K that are generally accessible via the Chowla–Selberg machinery rather than via Mahler measures of small Laurent polynomials. The two methodologies aim at the same final numbers but use disjoint formalisms.

For E_{lemn} specifically: $L(E_{\mathrm{lemn}}, 1) = \varpi/4$ [1, §11] is a Chowla–Selberg-style identity, not a Mahler-measure one.

Observation (What Mahler measures CAN say about G_K at $d = 2$). For $K = \mathbb{Q}(\sqrt{-2})$ (conductor of the canonical CM curve $E_{d=2}$ is $256 = 2^8$, also outside Boyd's tables), the Mahler-measure direction is similarly obstructed. The accessible Mahler-measure identities (Smyth, Boyd, Lawton) concern non-CM curves of small conductor; the entire CM family E_K for K in the h=1 atlas lies outside the standard Mahler-measure tabulation.

5. Open questions

The η-tower theorem closes one direction completely. Two natural follow-up programmes remain open:

E1. Higher-weight value algebra at the seven new fields (Conjecture 3.2). For each of $K \in \{\mathbb{Q}(\sqrt{-d}) : d \in \{2, 7, 11, 19, 43, 67, 163\}\}$, determine the algebra of values $\{E_{2k}(\tau_K) : k \geq 2\}$ explicitly. The natural framework is the level-N modular value algebra at the Atkin–Lehner fixed point.

E2. Mahler-measure analog for CM curves. Develop a "CM Mahler measure" replacement for Boyd's tables: a tabulation of Laurent polynomials $P(x, y)$ whose Mahler measures $m(P)$ equal Chowla–Selberg constants G_K (rather than L'-values of non-CM curves). Such a tabulation would close the gap between the two formalisms at the specific case of the conductor-32 obstruction.

References

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